

1. Briefly explain the meaning of the following terms:
 - a) Outlier
 - b) Trend
 - c) Turning point
 - d) Random shock
2. Describe four main components of time series data and demonstrate how each of them can be identified. (You may use graphs to demonstrate your answer).
3. Explain the effect of technological change on time series data.
4. The following data represents annual expenditure (RM'000) of a small company for the years 1966-1996. Analyse and comment the data set. (Graphs are useful to support your argument. Students should attempt to highlight the main characteristics of the data set).

Year	Expenditure (RM'000)
1966	24
1967	30
1968	36
1969	42
1970	50
1971	49
1972	23
1973	34
1974	53
1975	74
1976	87
1977	55
1978	93
1979	136
1980	211

Year	Expenditure (RM'000)
1981	113
1982	228
1983	321
1984	409
1985	480
1986	75
1987	46
1988	52
1989	76
1990	106
1991	95
1992	96
1993	101
1994	112
1995	140
1996	142

5. Identify the components that characterize the following time series representing quarterly fish output in Malaysia.

Year	Quarters	Tons	Year	Quarters	Tons
1990	1	211327	1994	1	229826
	2	248520		2	290660
	3	259806		3	291204
	4	231654		4	253895
1991	1	211310	1995	1	246057

	2	233831		2	306053
	3	248664		3	294712
	4	218128		4	261614
1992	1	245730	1996	1	249476
	2	246819		2	294621
	3	274028		3	313968
	4	256939		4	268624
1993	1	244513	1997	1	274919
	2	257018		2	209853
	3	285892		3	293518
	4	259927		4	290683

6. In simple term, explain the assumptions that can be made concerning the relationship between the time series components.
7. What do you understand by 'turning point' as commonly found in time series data? Explain and give suitable examples for your answer.
8. What do you understand by the term 'detrended series'? Briefly explain.
9. Briefly explain the meaning of 'classical decomposition method'.
10. Using data in Question 5, generate forecast values for all quarters in 1998.
11. In what way is multiplicative effect assumption different from the additive assumption?
12. Write the equation for a simple linear trend and explain all the symbols used.
13. How do you explain the differences between short-term and long-term memory effects?
14. Give two examples for the causes of random shock.
15. What can you say about the weights used in the calculations of moving average? Provide a simple example.
16. Why is it difficult to forecast the cyclical effect?
17. Given the following seasonal indices, what can you say about the time series?

Quarters	Q1	Q2	Q3	Q4
Seasonal indices	80	92	110	118

18. Why is seasonal effect important to forecasters?
19. What is the reason that makes it difficult to generate forecast values using the moving average method?
20. State and sketch two irregularities pertaining to the trend that you know of.